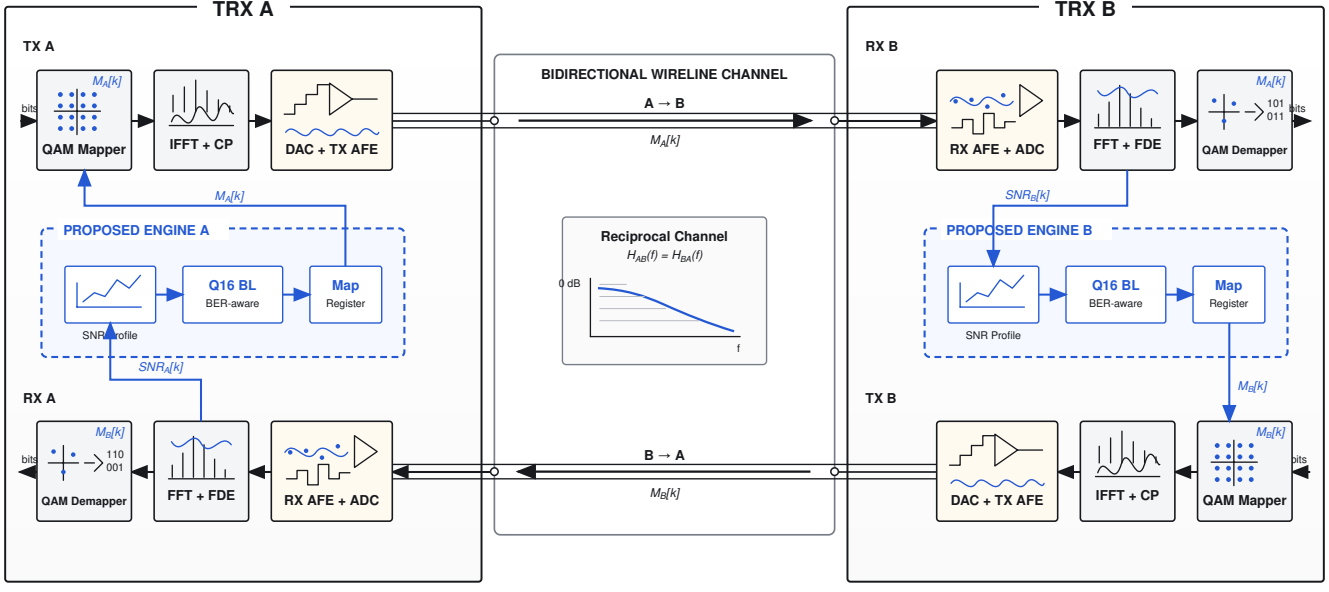
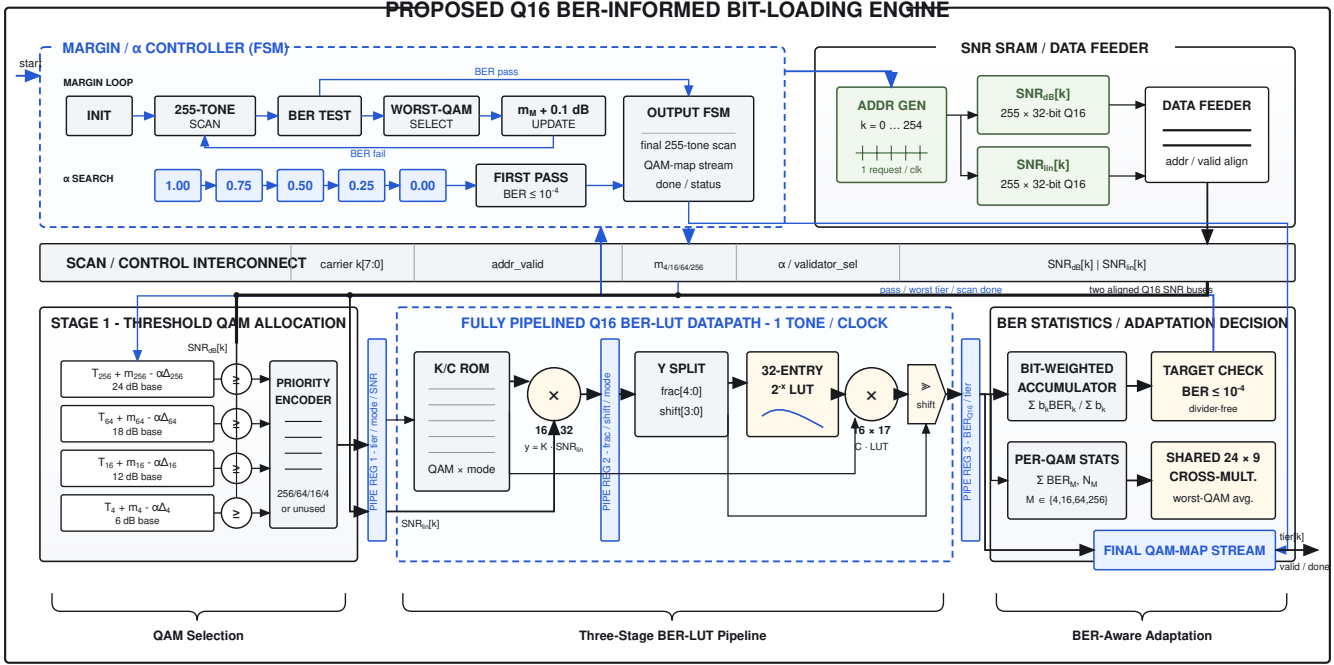

TRX A
TX A
bits
$M_A[k]$
QAM Mapper
IFFT + CP
DAC + TX AFE
$M_A[k]$
PROPOSED ENGINE A
SNR Profile
Q16 BL
BER-aware
Map
Register
$SNR_A[k]$
RX A
bits
$M_B[k]$
110
001
QAM Demapper
FFT + FDE
RX AFE + ADC
BIDIRECTIONAL WIRELINE CHANNEL
A → B
$M_A[k]$
Reciprocal Channel
$H_{AB}(f) = H_{BA}(f)$
0 dB
f
B → A
$M_B[k]$
TRX B
RX B
RX AFE + ADC
FFT + FDE
$M_A[k]$
101
011
QAM Demapper
bits
$SNR_B[k]$
PROPOSED ENGINE B
SNR Profile
Q16 BL
BER-aware
Map
Register
$M_B[k]$
TX B
DAC + TX AFE
IFFT + CP
$M_B[k]$
QAM Mapper
bits


PROPOSED Q16 BER-INFORMED BIT-LOADING ENGINE
MARGIN / α CONTROLLER (FSM)
start
MARGIN LOOP
INIT
255-TONE SCAN
BER TEST
WORST-QAM SELECT
m_M + 0.1 dB UPDATE
BER pass
BER fail
OUTPUT FSM
final 255-tone scan
QAM-map stream
done / status
α SEARCH
1.00
0.75
0.50
0.25
0.00
FIRST PASS
BER ≤ 10^-4
SNR SRAM / DATA FEEDER
ADDR GEN
k = 0 … 254
1 request / clk
SNR_dB[k]
255 × 32-bit Q16
SNR_lin[k]
255 × 32-bit Q16
DATA FEEDER
addr / valid align
SCAN / CONTROL INTERCONNECT
carrier k[7:0]
addr_valid
m_4/16/64/256
α / validator_sel
SNR_dB[k] | SNR_lin[k]
pass / worst tier / scan done
two aligned Q16 SNR buses
STAGE 1 - THRESHOLD QAM ALLOCATION
SNR_dB[k]
T_256 + m_256 - αΔ_256
24 dB base
T_64 + m_64 - αΔ_64
18 dB base
T_16 + m_16 - αΔ_16
12 dB base
T_4 + m_4 - αΔ_4
6 dB base
≥
PRIORITY ENCODER
256/64/16/4 or unused
PIPE REG 1 - tier / mode / SNR
SNR_lin[k]
FULLY PIPELINED Q16 BER-LUT DATAPATH - 1 TONE / CLOCK
K/C ROM
QAM × mode
×
16 × 32
y = K · SNR_lin
PIPE REG 2 - frac / shift / mode
Y SPLIT
frac[4:0]
shift[3:0]
32-ENTRY 2^x LUT
×
16 × 17
C · LUT
≫
shift
PIPE REG 3 - BER_Q16 / tier
BER STATISTICS / ADAPTATION DECISION
BIT-WEIGHTED ACCUMULATOR
Σ b_k BER_k / Σ b_k
TARGET CHECK
BER ≤ 10^-4
divider-free
PER-QAM STATS
Σ BER_M, N_M
M ∈ {4,16,64,256}
SHARED 24 × 9 CROSS-MULT.
worst-QAM avg.
FINAL QAM-MAP STREAM
tier[k]
valid / done
QAM Selection
Three-Stage BER-LUT Pipeline
BER-Aware Adaptation